

Speaker and Digit Representation based Voice OTP System

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Abstract—Voice OTP Authentication (VOA) is a two-way authentication framework to automatically authenticate the user by identifying the speaker and spoken One-Time Password (OTP). The work towards designing the VOA system is limited in the literature. This work proposed a novel framework, that uses state-of-the-art (SOTA) Emphasized Channel Attention, Propagation, and Aggregation Time Delayed Neural Network (ECAPA-TDNN) based x-vector as speaker representation and conformer-based digit representation to perform VOA. The performance of VOA using the pre-trained publicly available speaker and digit representation extractor is 78.92%. The performance is improved to 93.21% after fine-tuning the frameworks with the in-domain training data. Further, to see the feasibility of the framework for deployment, the inference time and memory requirement are calculated by deploying the framework in three devices: (1) high-end server, (2) laptop CPU, and (3) low-end edge device (odroid board). The inference times in all three devices are 0.068s, 0.33s, and 1.74s, respectively.

Index Terms—Voice OTP Authentication, Speaker Representation, Digit Representation

I. INTRODUCTION

With more people working from different places worldwide, there's a growing need for secure ways to let them access their accounts and data online. VOA is one such authentication system, offering a two-tier authentication process that validates the speaker's identity and the correctness of a spoken OTP. Technologies like Speaker Recognition (SR) and Digit Recognition (DR), are well-explored in the literature. However, the literature on VOA is limited. VOA is beneficial for people who find typing or using passwords tricky. It's cost-effective and easy to use, making it an excellent choice for various situations. Its applicability extends beyond the scope of online banking, encompassing scenarios such as remote workstation logins, attendance tracking systems, and various other applications. This study proposes a VOA system with two-factor authentication, and the flow of a use case is depicted in Figure 1. The system creates a numeric OTP, sends it to the user, and then confirms the user's identity and decodes the uttered OTP. The access is granted only if both speaker verification and OTP verification are successful.

In the early days, SR and DR methods were explored using various feature extraction and modeling techniques [4], [10]. Over time, these techniques have evolved, delving into the perception (e.g., Mel-Frequency Cepstral Coefficients (MFCC),

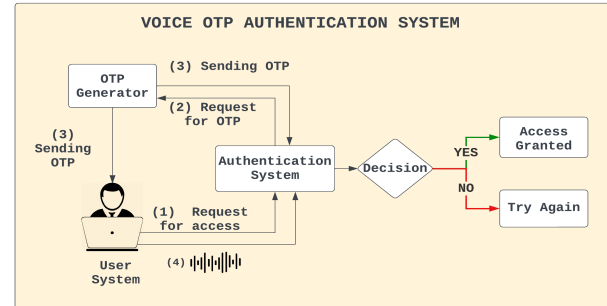


Fig. 1. Working of VOA System

Perceptual Linear Prediction (PLP)) and production (e.g., linear prediction residual) aspects of speech [8]–[10], [12], [13], [15]. The dynamics of formant locations play a crucial role in identifying speakers and digits [3], [10]. Among these methods, MFCC has demonstrated its effectiveness for both SR and DR tasks [3], [10]. However, the application of traditional approaches is limited to certain recording devices and environments. Nowadays, the evolution of deep learning-based frameworks can resolve the issue to some extent by training the deep learning frameworks with a huge amount of publicly available data. For speaker recognition, the most used techniques are Deep neural networks (DNN)-i-vector, d-vector, x-vector, and ECAPA-TDNN based x-vector [7]. Out of this, the performance of the ECAPA-TDNN-based x-vector is superior to the rest [7]. Similarly, the deep learning models like Deep Neural Network-Hidden Markov Model (DNN-HMM) [14], DeepSpeech2 [1], [2], and wav2vec-conformer models [6] are mostly used techniques to perform speech recognition tasks. Out of them, the wav2vec-conformer models perform comparatively better than the rest [6].

Our hypothesis posits that leveraging speaker representations from the ECAPA-TDNN model (trained on VoxCeleb) and digit representations from the conformer-based wav2vec (W2V) model (fine-tuned with Indian English) may be advantageous in the development of VOA systems. Further, evaluating the performance of VOA using the publicly available speaker and digit representation extractor, an in-domain fine-tuning strategy is explored to improve the VOA performance. Furthermore, the inference time and the run-time memory

requirements are also investigated to study the feasibility of deploying the VOA system with different end devices.

The rest of the paper is organized as follows: Section II is dedicated to the motivation section, which discusses the motivation for using the speaker and digit representation and in-domain fine-tuning. Next, in Section III, the proposed framework of the VOA system is addressed, including the proposed dataset, OTP generation, and the VOA system's main components. Subsequently, in Section IV, the experiment setup and the fine-tuning of the models are elaborated upon, comparing various parameters such as accuracy, time, and memory between the pre-trained and fine-tuned models. The final section, Section V, highlights how the performance of the VOA system has been enhanced through the utilization of domain data and discusses how the fine-tuning process has impacted other metrics.

II. MOTIVATION

This section starts the discussion with the distribution of t-distributed Stochastic Neighbor Embedding (t-SNE) for speaker and digit representations, before and after fine-tuning the models with speaker data. The objective is to assess whether distinct clustering patterns emerge in speaker representations from the ECAPA TDNN model (trained with the Voxceleb data and available at ¹) and digit representations from Wav2Vec Conformer model and to evaluate how the cluster distinction has improved after fine-tuning with the speaker data used for plotting in both cases.

In this t-SNE analysis, utterances from 10 speakers, each contributing 100 utterances, were employed to observe speaker discrimination. Speaker representations were derived by extracting filter bank features with a 25 msec frame size and 10 msec frameshift. These features were subsequently processed by the ECAPA-TDNN model, resulting in 192-dimensional speaker representations. These representations from all 10 speakers were then projected into 2 dimensions using t-SNE. Subsequently, the ECAPA-TDNN model underwent fine-tuning using the speaker data employed for plotting. Then, a similar t-SNE analysis was conducted on the speaker representations obtained from the fine-tuned ECAPA-TDNN model.

In Figure 2, the visual representation demonstrates that the speaker representations form distinct clusters corresponding to individual speakers with minimal overlap. This clustering phenomenon is particularly good in the case of the fine-tuned model, providing motivation for leveraging fine-tuned ECAPA-TDNN-based speaker representations for speaker identification tasks.

In the context of digit representation, utterances of digits 0 to 5 from one of the sessions are taken from 50 speakers for each digit. These audio samples were directly processed through a Wav2Vec model trained on Indian English data, resulting in dimensions $(1, x, 23)$, where the value of x varies for each individual audio sample. To standardize the dimensionality across all samples, the mean was computed along

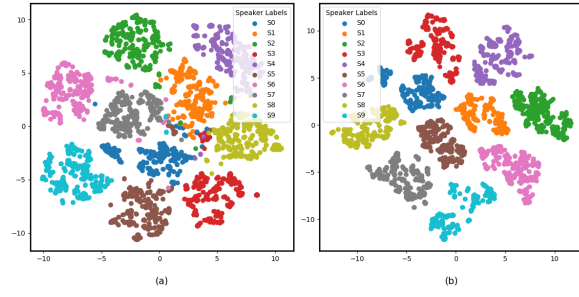


Fig. 2. t-SNE Visualization of Speaker representations from (a) pre-trained ECAPA-TDNN model and (b) fine-tuned ECAPA-TDNN model.

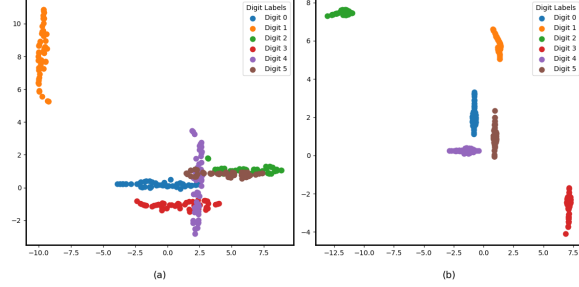


Fig. 3. t-SNE Visualization of Digit representations for 0-6 digits from (c) pre-trained wav2vec model and (d) fine-tuned wav2vec model.

$axis - 1$, leading to the transformation of all representations into a uniform format of $(1, 23)$. These representations are then used for t-SNE analysis. Likewise, the t-SNE analysis was conducted on the representations extracted after fine-tuning the Wav2Vec model using the identical data employed for plotting.

Figure 3 illustrates that digit representations exhibit separate clusters with notably high clustering quality in the fine-tuned model. These findings underscore the motivation to employ the fine-tuned Wav2Vec model for OTP identification tasks.

III. PROPOSED FRAME WORK

This study employed a dataset comprising audio samples collected inside the institute from 50 speakers, consisting of 47 males and 3 females. Each speaker's participation encompassed 4 sessions capturing different environments. In each session, the utterances of digits ranging from 0 to 9 are collected in one recording. From these recordings, separate audio samples for each digit from 0 to 9 are generated by segmenting the audio, which are then used to produce OTP datasets of different lengths. Throughout the paper, this dataset is referred to as *DigitUtter* dataset and available at ².

Moreover, OTPs with lengths ranging from 1 to 4 were generated from the *DigitUtter* dataset for each speaker and each session by merging the audios of various digits to encompass all possible N -digit OTPs. Throughout the paper, the number of digits in the OTP is referred to as N . For instance, for two-digit OTPs, it is referred to as the 2-digit-OTP dataset. Table I details the sample distribution among

¹https://catalog.ngc.nvidia.com/orgs/nvidia/teams/nemo/models/ecapa_tdn

²<https://github.com/mcqueen444/DigitUtter-IITDH-dataset>

speakers within each OTP dataset. The detailed of dataset can be found in our earlier study reported in [11].

Figure 4 represents the overall structure of the proposed VOA system consisting of three main components: (1) Speaker Identification (SI), (2) OTP Identification (OI), and (3) Decision Logic (DL). Further elaboration on each component is presented in the following subsections.

TABLE I
SUMMARY OF OTP DATASETS

N-digit-OTP Dataset	1	2	3	4
Utterances/Speaker/Session	10	100	1000	10000
Total Training Utterances/Speaker	30	300	3000	30000
Total Testing Utterances/Speaker	10	100	1000	10000

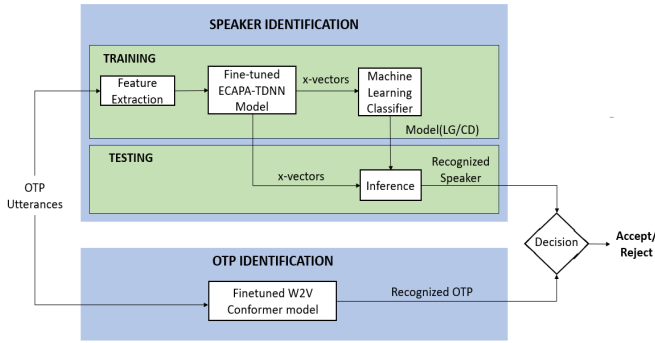


Fig. 4. Block diagram for Voice OTP Authentication system. LR and CD denote Logistic Regression and Cosine Distance, respectively

A. Speaker Identification

The SI task involves utilizing speaker representations derived from the fine-tuned ECAPA-TDNN model. This model takes filter bank features as input extracted from speech signals and is specifically trained for speaker classification. It is a sophisticated model consisting of TDNN layers, a temporal pooling layer, and fully connected layers, operating at both the frame and utterance levels. The ECAPA-TDNN model, accessible through the NVIDIA NeMo toolkit³, has undergone pre-training on the VoxCeleb-1 dataset and two additional datasets, encompassing a substantial 7205 speakers and numerous hours of speech data. The model is subsequently fine-tuned using the *DigitUtter* dataset to tailor it for the SI task.

Following the extraction of filter bank features from each spoken utterance, this fine-tuned ECAPA-TDNN model generates speaker representations known as *x-vectors*. These *x-vectors* are used in two distinct approaches for the speaker identification task first, in a model-based approach that involves training of Logistic Regression (LR), and second, in a model-free approach that relies on a straightforward Cosine Distance (CD) based comparison method, utilizing the fine-tuned ECAPA-TDNN model.

³https://catalog.ngc.nvidia.com/orgs/nvidia/teams/nemo/models/ecapa_tdn

B. OTP Identification

The process of OTP identification includes decoding OTP utterances using an Automatic Speech Recognition (ASR) model specifically trained for Indian English. This ASR model, the W2V-based conformer model, has been developed and open-sourced by the Vakayansh team [5] for performing OTP decoding. The training of the W2V model typically involves two stages: (1) pre-training and (2) fine-tuning.

In the pre-training stage, the model is trained using unlabeled speech data collected from a diverse range of 39 Indian languages, totalling approximately 35,000 hours of data. Following this phase, the pre-trained network undergoes fine-tuning using 700 hours of labelled Indian English speech data to specialize it further. The fine-tuned model can be found at ⁴.

Furthermore, this fine-tuned model is subject to an additional round of fine-tuning to adapt it to the specific vocabulary used in the *DigitUtter* dataset, which comprises OTP digit utterances. This entails further customization of the model through fine-tuning with OTP utterances of length spanning from 1 to 4 digits for all 50 speakers in the dataset. Subsequently, this finely tuned model is employed for OTP decoding during the testing phase of the framework.

C. Decision Logic

This module in the VOA system integrates outputs from the SI and OI modules to reach a final decision. During this phase, the identified speaker and the decoded OTP are cross-verified against the claimant's identity and the generated OTP. Authentication is deemed successful only when both verifications yield positive results; otherwise, it is rejected.

IV. EXPERIMENTAL SETUP, RESULTS AND DISCUSSION

In this section, the performance of the proposed system is assessed and discussed through the following series of experiments.

- 1) First, the ECAPA-TDNN model will be fine-tuned, with model-based and model-free approaches being applied for SI tasks and fine-tuning the Wav2Vec Conformer model for OI tasks. OTP datasets of different lengths (ranging from 1 to 4) from the *DigitUtter* dataset will be employed for the fine-tuning.
- 2) Next, the overall accuracy of the fine-tuned VOA system is calculated and compared to results derived directly from pre-trained models.
- 3) Finally, the performance of the integrated VOA system on various end devices is evaluated, with metrics such as accuracy, inference time, and memory usage.

This study involved three different devices with varying performance capabilities to investigate performance differences and evaluate their potential for practical deployment.

⁴<https://github.com/Open-Speech-EkStep/vakayansh-models.git>

TABLE II
PRETRAINED (PT) AND FINE-TUNED (FT) ACCURACY (%) FOR SI WITH LR, CD APPROACHES, AND FOR OI TASKS.

N	SI				OI		
	PT		FT		No. of Samples	PT	FT
	LR	CD	LR	CD			
1	66.27	53.09	82.16	47.94	500	50.70	81.36
2	87.29	64.69	90.86	88.79	5000	74.58	94.58
3	94.23	67.15	95.97	88.05	50000	80.18	95.24
4	96.75	74.45	97.28	92.61	50000	81.16	95.66

- **Server CPU (SC)** : 24-Core W-2265 CPU@3.50 GHz.
- **Desktop CPU (LC)**:8-Core i7-8550U CPU@1.80 GHz.
- **Odroid Device (OD)**:4-core CPU@1.80 GHz and 2-core CPU@1.90 GHz.

The ECAPA TDNN model from the NVIDIA NeMo toolkit was fine-tuned with each 1 to 4 OTP dataset for the SI task following a comprehensive configuration. The preprocessing steps included normalizing the audio data per feature, employing a window size of 0.025 seconds, and utilizing a sample rate of 16,000 Hz. The encoder architecture was based on the ECAPA TDNN framework. The decoder, tasked with identifying speakers, incorporated an attention-based pooling mode, and angular softmax loss was used during training with a scale of 30 and a margin of 0.2. Optimization employed the SGD optimizer with a learning rate of 0.08. This fine-tuning process aimed to adapt the ECAPA TDNN model to the specific SI task, enabling it to distinguish among the speakers within the dataset effectively.

From the fine-tuned ECAPA TDNN model, x-vectors are extracted by processing the filterbank features of utterances from all N-digit OTP datasets and sessions. These x-vectors are then used to train individual LR classifiers for each OTP dataset. During the testing phase, audio filterbanks are passed through the fine-tuned ECAPA- TDNN model. The resulting x-vectors are fed into the corresponding trained LR model for speaker identification.

In the CD-based approach, speaker-specific mean vectors are computed from all x-vectors within the training data corresponding to each speaker. These mean vectors are preserved as representations for each speaker. The test x-vector is compared to all the stored speaker representations utilizing the CD metric in the testing phase. The identified speaker is determined by selecting the speaker representation with the maximum cosine distance score. Detailed results for both approaches with pre-trained and fine-tuned models are presented in Table II.

Vakyansh is a speech recognition toolkit powered by wav2vec and trained in multiple Indian languages, delivering state-of-the-art performance. As indicated in table II, the initial results for the OI task when using the model directly were unsatisfactory. To enhance performance, our hypothesis was to

TABLE III
ACCURACY (%) FOR SI, OI, COMBINED ACCURACY BEFORE AND AFTER FINE-TUNING.

N	Pretrained			fine-tuned		
	SI	OI	Combined	SI	OI	Combined
1	66.27	50.70	32.87	82.16	81.36	68.73
2	87.29	74.58	66.14	90.86	94.57	86.46
3	94.23	80.18	75.89	95.97	95.24	91.89
4	96.75	81.16	78.92	97.28	95.66	93.21

assess whether fine-tuning the model with the DigitUtter dataset would improve accuracy. For this fine-tuning process, identical training and testing data that were previously employed for fine-tuning the ECAPA-TDNN model were utilized.

In this fine-tuning process, the training data is used to generate the vocabulary, which consists of digits, as it aligns with OTPs. Previously, this vocabulary comprised text-related terms from the large dataset on which the model was initially trained. During the fine-tuning of this speech recognition model, the model is initially restored from the previous checkpoint, serving as the foundational starting point. The training environment is configured, and this configuration is customized for training machine learning models, allowing for the utilization of all available GPUs, setting a maximum of 30 training epochs, and conducting validation checks once per epoch. The training strategy employs Distributed Data-Parallel (DDP) and facilitates gradient accumulation. The model, refined through the fine-tuning process, is then assessed using the test data, and the outcomes of testing it with OTP utterances ranging from 1 to 4 digits are presented in table II. By examining the results, it becomes evident that fine-tuning the model with the dedicated dataset for the OI task has significantly boosted its performance. Additionally, it has been observed that employing 5,00,000 samples for the $N = 4$ case in the OI task yielded similar results to those obtained using 50,000 samples. Hence, this observation has enabled the decrease in fine-tuning expenses for $N = 4$ by utilizing a dataset comprising only 50,000 samples.

The combined accuracy of the VOA system is computed by incorporating the LR approach with a fine-tuned ECAPA TDNN model for SI task and fine-tuned Wav2Vec model for the OI task. The combined results in terms of accuracy for pre-trained and fine-tuned models in the context of SI, OI, and their combination are presented in Table III. Furthermore, the combined outcomes represent the intersection of SI and OI tasks, resulting in lower accuracy compared to each individual task.

The following aspect involves assessing the time consumption across different devices while comparing the model-free CD and model-based LR approaches for the fine-tuned models. Table IV presented, showcasing the inference time of the fine-tuned models on various devices. These results are based on a small sample of OTP utterances.

When considering inference across various devices and after thoroughly examining the experimental results in Table IV, it

TABLE IV
INFERENCE TIME IN SECONDS FOR VOA SYSTEM WITH FINE-TUNED MODELS

N	SC		LC		OD	
	LR	CD	LR	CD	LR	CD
1	0.068	0.052	0.330	0.251	1.743	1.641
2	0.072	0.076	0.421	0.362	2.223	2.121
3	0.081	0.059	0.564	0.485	2.781	2.700
4	0.111	0.077	0.637	0.723	4.504	3.083

TABLE V
MEMORY CONSUMPTION (MiB) INCREMENT AT EACH STEP OF THE SPEAKER VERIFICATION PROCESS RESPECTIVELY FOR PRE TRAINED AND FINE-TUNED

Step	Pretrained	fine-tuned
Importing modules	520 - 525	630 - 660
ECAPA TDNN model loading	1930 - 1940	1520 - 1530
Embedding extraction	3626 - 3628	2730 - 2740
LR model loading + predicting	3626 - 3628	2732 - 2742
Loading Wav2Vec	4338 - 4340	4490 - 4520
Transcription	4340 - 4342	4492 - 4522

becomes clear that the SC demonstrates the shortest inference time. It performs well, achieving an average inference time of approximately 0.081 seconds for a 3-digit OTP. In contrast, the OD requires significantly more time for inference, with its inference time measuring at 2.78 seconds. It's worth noting that the accuracy remains consistent across all devices, as the same approaches are applied uniformly.

In pursuit of implementing the VOA system on a resource-constrained device, the primary focus was on the reduction of both computation time and memory usage. To address concerns related to memory consumption, a comprehensive analysis of runtime memory usage across various stages of both SI and OI tasks was performed.

A detailed analysis of memory consumption was conducted throughout the VOA process. Modules that exhibited higher memory usage included the module for importation, ECAPA TDNN model loading, embedding extraction, and Vakyansh model loading (as referenced in Table V). The reported memory consumption results are specific to our model, which was trained and evaluated using four-digit utterances. During the VOA process, the actual prediction stage required less than 3 MiB of memory. A slight increase in memory consumption is observed after fine-tuning. Future efforts will focus on reducing this memory usage to accommodate the constraints of an edge device.

V. CONCLUSION

In summary, this study highlights the importance of utilizing speaker and digit representations obtained from the ECAPA-TDNN and W2V-Conformer models in developing the VOA system. The SI component demonstrates an accuracy of 96.75% and 97.28% when utilizing representations from the pre-trained and fine-tuned ECAPA-TDNN models, respectively. Additionally, the OI component showcases an improvement in accuracy from 81.16% to 95.66% through the fine-tuning of the W2V-Conformer model. Overall, the fine-tuning process leads to an improvement in the VOA system accuracy from 78.92% to 93.21%.

In the future, further techniques may be explored to decrease the inference time and memory consumption while also boosting the models' accuracy, making them suitable for use in low-resource devices.

VI. ACKNOWLEDGEMENT

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